

Spacetime Density Kernel and the DSSYK Two-Point Function

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1 Setup: the Spacetime Density Kernel

Given a Hilbert space $\mathcal{H}$ with density matrix ρ , time evolution $U_t = e^{-iHt}$, and a bipartition $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ with matrix unit bases $\hat{E}_{ij}^A = |i\rangle\langle j| \otimes \mathbf{1}_B$ and $\hat{E}_{kl}^B = \mathbf{1}_A \otimes |k\rangle\langle l|$, the *spacetime density kernel* (SDK) coefficients are defined by

$$C_{ij;kl}(t) = \text{Tr} \left[\rho \hat{E}_{ij}^A U_t^\dagger \hat{E}_{kl}^B U_t \right]. \quad (1)$$

The SDK encodes two-time correlators via

$$\mathcal{L}_t(O_A, O_B) = \text{Tr} \left[\rho O_A U_t^\dagger O_B U_t \right] = \sum_{i,j,k,l} C_{ij;kl}(t) O_{ij}^{(A)} O_{kl}^{(B)}, \quad (2)$$

where $O_{ij}^{(A)} = \langle i|O_A|j\rangle$ and $O_{kl}^{(B)} = \langle k|O_B|l\rangle$ are the matrix elements in the chosen basis.

2 Single-Sided DSSYK

2.1 Energy basis and overlaps

The single-sided DSSYK Hilbert space is spanned by chord number states $|n\rangle$, $n = 0, 1, 2, \dots$. We take $\rho = |0\rangle\langle 0|$. The continuous energy eigenstates $|\theta\rangle$ satisfy the resolution of identity

$$\mathbf{1} = \int_0^\pi \frac{d\theta}{2\pi} \mu(\theta) |\theta\rangle\langle\theta|, \quad \mu(\theta) = (q, e^{\pm 2i\theta}; q)_\infty, \quad (3)$$

with the inner product

$$\langle\theta|\phi\rangle = \frac{2\pi \delta(\theta - \phi)}{\mu(\theta)}. \quad (4)$$

The overlaps with chord basis states are

$$\langle\theta|n\rangle = \frac{H_n(\cos\theta|q)}{\sqrt{(q; q)_\infty}}, \quad (5)$$

where $H_n(x|q)$ are the continuous q -Hermite polynomials, $H_0 = 1$. The energy is parametrized by

$$E(\theta) = -\frac{2J \cos\theta}{\sqrt{\lambda(1-q)}}. \quad (6)$$

2.2 SDK coefficients

Inserting two resolutions of identity (3) into (1) and using $\rho = |0\rangle\langle 0|$ gives

$$C_{ij;kl}(t) = \int \frac{d\theta d\phi}{(2\pi)^2} \mu(\theta)\mu(\phi) e^{-it(E(\theta)-E(\phi))} \langle 0|\hat{E}_{ij}^A|\theta\rangle \langle \theta|\hat{E}_{kl}^B|\phi\rangle \langle \phi|0\rangle. \quad (7)$$

Evaluating each matrix element,

$$\langle 0|\hat{E}_{ij}^A|\theta\rangle = \langle 0|i\rangle\langle j|\theta\rangle = \delta_{0i} \frac{H_j(\cos\theta|q)}{\sqrt{(q;q)_\infty}}, \quad (8)$$

$$\langle \theta|\hat{E}_{kl}^B|\phi\rangle = \langle \theta|k\rangle\langle l|\phi\rangle = \frac{H_k(\cos\theta|q) H_l(\cos\phi|q)}{(q;q)_\infty}, \quad (9)$$

$$\langle \phi|0\rangle = \frac{H_0(\cos\phi|q)}{\sqrt{(q;q)_\infty}} = \frac{1}{\sqrt{(q;q)_\infty}}. \quad (10)$$

Substituting back into (7) and absorbing overall factors of $(q;q)_\infty$ into a proportionality constant,

$$C_{0j;kl}(t) \propto \delta_{0i} \int \frac{d\theta d\phi}{(2\pi)^2} \mu(\theta)\mu(\phi) e^{-it(E(\theta)-E(\phi))} \frac{H_j(\cos\theta|q)^*}{\sqrt{(q;q)_j}} \frac{H_k(\cos\theta|q)}{\sqrt{(q;q)_k}} \frac{H_l(\cos\phi|q)}{\sqrt{(q;q)_l}}. \quad (11)$$

3 Contraction with the Matter Operator

3.1 The operator and its matrix elements

The physical matter operator $\mathcal{O}_\Delta$ in DSSYK is diagonal in the chord basis:

$$O_{ij} = \langle i|\mathcal{O}_\Delta|j\rangle = \delta_{ij} \tilde{q}^i, \quad \tilde{q} = q^\Delta. \quad (12)$$

In the chord diagram language, each crossing of an M -string with an H -string contributes a factor $\tilde{q}$, so $\mathcal{O}_\Delta$ is represented by the diagonal matrix $S = \text{diag}(1, \tilde{q}, \tilde{q}^2, \dots)$.

The NV physical operator is a product of left and right local operators,

$$\mathcal{O}_\Delta^{\text{phys}}(\tau) = \int dt \mathcal{O}_{1-\Delta}^L(t) \mathcal{O}_\Delta^R(-t + \tau), \quad (13)$$

so the full contraction requires *two* operator coefficient matrices: $O_{ij}^L = \delta_{ij}(\tilde{q}')^i$ with $\tilde{q}' = q^{1-\Delta}$, and $O_{kl}^R = \delta_{kl}\tilde{q}^k$ with $\tilde{q} = q^\Delta$.

3.2 Applying the Poisson kernel identity

Contracting (11) with $O_{ij}^L O_{kl}^R$ and using δ_{ij} , δ_{0i} , δ_{kl} , the $i = j = 0$ constraint gives $(\tilde{q}')^0 = 1$, and the sum over $k = l$ remains:

$$\begin{aligned} \mathcal{L}_\tau(\mathcal{O}_\Delta^{\text{phys}}, \mathcal{O}_\Delta^{\text{phys}}) &\propto \int \frac{d\theta d\phi}{(2\pi)^2} \mu(\theta)\mu(\phi) e^{-\tau(E(\theta)-E(\phi))} \left(\sum_j \frac{H_j(\cos\theta|q) H_j(\cos\phi|q)}{(q;q)_j} (\tilde{q}')^j \right) \times \\ &\times \left(\sum_k \frac{H_k(\cos\theta|q) H_k(\cos\phi|q)}{(q;q)_k} \tilde{q}^k \right). \end{aligned} \quad (14)$$

Each sum is evaluated using the Poisson kernel for the continuous q -Hermite polynomials [1],

$$\sum_{k=0}^{\infty} \frac{H_k(\cos \theta|q) H_k(\cos \phi|q)}{(q; q)_k} t^k = \frac{(t^2; q)_{\infty}}{(t e^{i(\pm\theta \pm \phi)}; q)_{\infty}}, \quad (15)$$

where the right-hand side is shorthand for the product over all four sign choices,

$$(t e^{i(\pm\theta \pm \phi)}; q)_{\infty} = (t e^{i(\theta + \phi)}; q)_{\infty} (t e^{i(\theta - \phi)}; q)_{\infty} (t e^{-i(\theta - \phi)}; q)_{\infty} (t e^{-i(\theta + \phi)}; q)_{\infty}. \quad (16)$$

Here t is the generating parameter; the left side is a formal power series in t whose coefficients are the products $H_k(\cos \theta|q) H_k(\cos \phi|q)/(q; q)_k$. In our context $t = \tilde{q} = q^{\Delta}$ encodes the scaling dimension of the operator.

Applying (15) to both sums in (14) with $t = \tilde{q} = q^{\Delta}$ and $t' = \tilde{q}' = q^{1-\Delta}$ respectively,

$$\mathcal{L}_{\tau} \propto \int \frac{d\theta d\phi}{(2\pi)^2} \mu(\theta) \mu(\phi) e^{-\tau(E(\theta) - E(\phi))} \frac{((\tilde{q}')^2; q)_{\infty}}{(\tilde{q}' e^{i(\pm\theta \pm \phi)}; q)_{\infty}} \cdot \frac{(\tilde{q}^2; q)_{\infty}}{(\tilde{q} e^{i(\pm\theta \pm \phi)}; q)_{\infty}}. \quad (17)$$

3.3 Converting to q -Gamma functions

Using the definition of the q -Gamma function,

$$\Gamma_q(x) = (1 - q)^{1-x} \frac{(q; q)_{\infty}}{(q^x; q)_{\infty}}, \quad (18)$$

and identifying $q^{is} = e^{i\theta}$, each q -Pochhammer factor converts as

$$\frac{1}{(\tilde{q} e^{i(\pm\theta \pm \phi)}; q)_{\infty}} = \frac{1}{\prod_{\epsilon_1, \epsilon_2 = \pm} (q^{\Delta + i\epsilon_1 s_1 + i\epsilon_2 s_0}; q)_{\infty}} \propto \Gamma_q(\Delta \pm i s_0 \pm i s_1), \quad (19)$$

and analogously for the $1 - \Delta$ factor. The numerator factors give $(\tilde{q}^2; q)_{\infty} = (q^{2\Delta}; q)_{\infty} \propto 1/\Gamma_q(2\Delta)$, and $((\tilde{q}')^2; q)_{\infty} \propto 1/\Gamma_q(2 - 2\Delta)$.

The Askey-Wilson measure converts as

$$\mu(\theta) = (q, e^{\pm 2i\theta}; q)_{\infty} \propto \frac{(q; q)_{\infty}^3}{\Gamma_q(\pm 2i s_1)}. \quad (20)$$

3.4 Collapsing the ϕ integral via the choice of ρ

To match the NV correlator, we must replace $\rho = |0\rangle\langle 0|$ by

$$\rho = |E_0\rangle\langle E_0|, \quad |E_0\rangle = |\theta_0\rangle, \quad (21)$$

where $|E_0\rangle = |\Psi_{\text{ds}}\rangle$ is the maximal entropy state, an energy eigenstate with spectral parameter s_0 and $E_0 = 0$. Since $|E_0\rangle$ is an energy eigenstate, the overlap with the chord basis is

$$\langle E_0|n\rangle\langle E_0|n\rangle = \langle \theta_0|n\rangle\langle \theta_0|n\rangle = \frac{H_n(\cos \theta_0 | q)}{\sqrt{(q; q)_{\infty}}}, \quad (22)$$

and in particular $\langle E_0|0|E_0|0\rangle = 1/\sqrt{(q;q)_\infty}$, which is no longer δ_{0i} . Crucially, with $\rho = |E_0\rangle\langle E_0|$ the SDK integral (7) becomes

$$C_{ij;kl}(t) = \int \frac{d\theta d\phi}{(2\pi)^2} \mu(\theta)\mu(\phi) e^{-it(E(\theta)-E(\phi))} \langle E_0| \hat{E}_{ij}^A |\theta\rangle \langle \theta| \hat{E}_{kl}^B |\phi\rangle \langle \phi|E_0|\phi|E_0\rangle. \quad (23)$$

Now $\langle \phi|E_0|\phi|E_0\rangle = \langle \phi|\theta_0|\phi|\theta_0\rangle = 2\pi\delta(\phi - \theta_0)/\mu(\phi)$ by the inner product (4). The $\delta(\phi - \theta_0)$ collapses the ϕ integral immediately, with the $1/\mu(\phi)$ canceling one power of $\mu(\phi)$ from the measure, leaving a single integral over θ :

$$C_{ij;kl}(t)|_{\rho=|E_0\rangle\langle E_0|} \propto \int \frac{d\theta}{2\pi} \mu(\theta) e^{-it(E(\theta)-E(\theta_0))} \langle E_0| \hat{E}_{ij}^A |\theta\rangle \langle \theta| \hat{E}_{kl}^B |\theta_0\rangle. \quad (24)$$

The phase $e^{+itE(\theta_0)}$ is a constant overall factor (since s_0 is fixed) and can be absorbed into normalization. The contraction with operator coefficients then proceeds exactly as in Section 5.2–5.3, with ϕ replaced everywhere by the fixed value θ_0 . Both applications of the Poisson kernel (15) now involve arguments $(\cos \theta, \cos \theta_0)$ rather than $(\cos \theta, \cos \phi)$, and the result is the single integral

$$\mathcal{L}_\tau(\mathcal{O}_\Delta^{\text{phys}}, \mathcal{O}_\Delta^{\text{phys}}) \propto \int_0^\pi \frac{d\theta}{2\pi} \mu(\theta) e^{-\tau E(\theta)} \frac{\Gamma_q(\Delta \pm is_0 \pm is_1) \Gamma_q(1 - \Delta \pm is_0 \pm is_1)}{\Gamma_q(\pm 2is_1) \Gamma_q(2\Delta) \Gamma_q(2 - 2\Delta)}, \quad (25)$$

where s_1 is the spectral parameter corresponding to θ , and s_0 is the fixed spectral parameter of the external state $|E_0\rangle$.

4 Comparison with the NV Two-Point Function

Equation (25) is precisely the DSSYK two-point function of Narovlansky and Verlinde [2] (eq. (11) of [3]):

$$G_\Delta(\tau_1) = \langle E_0|\mathcal{O}_\Delta(\tau)\mathcal{O}_\Delta(0)|E_0\rangle = \int ds_1 e^{-\tau_1 E(s_1)} \frac{\Gamma_q(\Delta \pm is_0 \pm is_1) \Gamma_q(1 - \Delta \pm is_0 \pm is_1)}{\Gamma_q(\pm 2is_1) \Gamma_q(2\Delta) \Gamma_q(2 - 2\Delta)}. \quad (26)$$

Using the identity (eq. (12) of [3])

$$\Gamma_q(x) \Gamma_q(1 - x) = iq^{1/8}(1 - q)(q; q)_\infty^3 \frac{q^{x/2}}{\vartheta_1\left(\frac{i\lambda}{2\pi}x, q\right)}, \quad (27)$$

each pair $\Gamma_q(\Delta \pm is_0 \pm is_1) \Gamma_q(1 - \Delta \pm is_0 \pm is_1)$ in (26) can be rewritten in terms of Jacobi theta functions. The spectral density becomes a single theta function,

$$\rho(s) = \frac{1}{\Gamma_q(\pm 2is)} \propto \vartheta_1\left(\frac{\lambda}{\pi}s, q\right), \quad (28)$$

and the full two-point function takes the form

$$G_\Delta(\tau_1) = \int dE(s_1) e^{-\tau_1 E(s_1)} \frac{\vartheta_1\left(\frac{i\lambda}{\pi}\Delta, q\right) \prod_a \vartheta_1\left(\frac{\lambda}{\pi}s_a, q\right)}{\prod_a \vartheta_1\left(\frac{\lambda}{2\pi}(i\Delta \pm s_0 \pm s_1), q\right)}, \quad (29)$$

with $a \in \{0, 1\}$. The equivalence between (26) and (29) follows from the Jacobi triple product identity, which relates each q -Pochhammer factor $(e^{2i\theta}; q)_\infty$ to $\vartheta_1(\theta/\pi, q)$.

References

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